Large **forest fire spread prediction**: data and computational science

T Artés, A Cortés, T Margalef - Procedia Computer Science, 2016 - Elsevier

… Therefore, a **model** must be introduced to estimate the wind along the terrain. … the **prediction model** works when coupling both models wind field **model** and **forest fire propagation model**. …

☆ Save 99 Cite Cited by 15 Related articles All 6 versions

Modeling **forest fire spread** using machine learning-based cellular automata in a GIS environment

Y Xu, D Li, H Ma, R Lin, F Zhang - Forests, 2022 - mdpi.com

… In this paper, we propose a machine learning-based **forest fire spread model** in which the traditional cellular automata framework is combined with LSSVM (least squares support vector …

☆ Save 99 Cite Cited by 30 Related articles All 6 versions »

Atmospheric turbulence and wildland fires: a **review**

WE Heilman - International journal of **wildland fire**, 2023 - CSIRO Publishing

… Palmer also presented a conceptual **model** to explain a spiralling turbulent twin-vortex … high-resolution simulations of ABL turbulence **dynamics** during **wildland fire** events are needed to …

☆ Save 99 Cite Cited by 12 Related articles All 8 versions

A **simulation** of probabilistic **wildfire** risk components for the continental United States

MA Finney, CW McHugh, IC Grenfell, KL Riley… - … Research and Risk …, 2011 - Springer

… To make the fire growth **simulation** efficient, fire **behavior** for the entire landscape was pre-processed for all combinations of moisture (three percentile categories: 80th, 90th, and 97th …

☆ Save 99 Cite Cited by 487 Related articles All 23 versions

Advancements in **forest fire** prevention: A comprehensive **survey**

F Carta, C Zidda, M Putzu, D Loru, M Anedda, D Giusto - Sensors, 2023 - mdpi.com

… [118] introduced a **wildfire spread model** implementing an … fire **simulation** models were conducted on a real **wildfire** in Getty, California, USA. The results demonstrate that the IGN **model** …

☆ Save 99 Cite Cited by 123 Related articles All 12 versions »

Adaptive **wildfire spread prediction** for complex terrain: modeling the effectiveness of sprinkler systems

J Kim, J Ahn, J Kang - Fire Ecology, 2024 - Springer

… of a comprehensive CFD **model** in mitigating **wildfire spread** using sprinkler systems as an … in **wildfire** within 10 h of ignition, significant influence of wind speed on **spread** patterns, and a …

☆ Save 99 Cite Cited by 5 Related articles All 8 versions

Comparing Accuracy of **Wildfire Spread Prediction** Models under Different Data Deficiency Conditions

J Zhou, W Jiang, F Wang, Y Qiao, Q Meng - Fire, 2024 - mdpi.com

… the **prediction** accuracy of existing **wildfire spread** models and … **behavior** yet identified. This study aims to simulate the impact of data scarcity on the accuracy of **wildfire spread prediction** …

☆ Save 99 Cite Cited by 1 Related articles All 4 versions »

… data-driven simulations of **wildfire spread**–Part I: Reduced-cost Ensemble Kalman Filter based on a Polynomial Chaos surrogate **model** for parameter estimation

MC Rochoux, S Ricci, D Lucor… - Natural Hazards and …, 2014 - nhess.copernicus.org

… fire **spread model**; FIREFLY could readily be replaced by any other front-tracking **wildfire spread** simulator, for instance FARSITE, FOREFIRE, PROMETHEUS or PHOENIX RapidFire. …

☆ Save 99 Cite Cited by 129 Related articles All 16 versions »

Simulating **wildfire spread** based on continuous time series remote sensing images and cellular automata

H Zhuang, N Liu, X Xie, X Xu, M Li… - … of **Wildland Fire**, 2025 - CSIRO Publishing

… CA **model** is proposed to accurately simulate **wildfire spread** … a **simulation** system for predicting **wildfire propagation** that … to this paper during the peer **review** process. All authors hereby …

☆ Save 99 Cite Related articles All 2 versions

Review of **wildfire** management techniques—Part I: Causes, prevention, detection, suppression, and data analytics

S Jazebi, F De Leon, A Nelson - IEEE Transactions on Power …, 2019 - ieeexplore.ieee.org

… in consecutive frames over time for remote monitoring of **wildfire spread**. In [45], a … **forest fire** with low consumption of energy. The fire **spread model** in [46] can produce fire **propagation** …

☆ Save 99 Cite Cited by 140 Related articles All 2 versions